Test Project

Competitor Information Document

Autonomous Mobile Robotics

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1. Introduction

The primary focus of the Emirates Skills 2026 Autonomous Mobile Robotics CID is to outline the requirements for Competitors to assemble/prepare/manage/repair robots to interpret multiple environments and tasks in various environments that will be introduced onsite at the competition. It is expected that robots will take the environmental information they have gathered and use this information to manage both their overall mobility and target object-handling experiences.

Several sectors are experiencing a surge in automation and robotics, leading to an increased demand for mobile robotic technicians and engineers who can develop and maintain robotics systems in this industry. As part of the Emirates Skills 2026 competition, participants are tasked with designing, constructing, and operating an Autonomous Mobile Robot that meets all performance requirements. To accomplish this, Competitors must carefully analyze the information presented in this document to identify the primary set of performance requirements for the Mobile Robot. The Autonomous Mobile Robotics skill at Emirates Skills 2026 will introduce various modules that target the specific skills of a Mobile Robotics Technician or Engineer.

The Robot will be able to function autonomously in multiple environments. The appendices present sample environments and guidelines that can be used for preparation purposes. For example, the Robot can manipulate objects of varying size and weight, from various locations on a course and deposit them in specified positions.

In addition, Competitors will be expected to manage their robot's mobility in an unknown environment. This experience will require the robots to demonstrate their capability to complete a navigation and localization task.

2. Description of project and tasks

2.1 Main Test Project

Teams of two Competitors are required to design and build a mobile robot that will efficiently operate in custom environments that represent real-world scenarios or tasks.

The main Test Project is broken down into various modules. Competitors are expected to identify the Primary Set of Mobile Robot Performance Requirements through analysis of the information provided in this document:

Robots are required to move mostly autonomously throughout any performance evaluation environment,
Robots are required to Take control of the various target objects and move them to within the performance environment.

Note: The Primary Task that Competitors must complete on C-1 is to prepare the team's laptops and complete the re-assembly of their robot (if required) in time to make it available to complete tasks as required on C1.

3. Instructions to the Competitor

3.1 Competition Requirement

Each Team will bring to the competition **1** Studica Robot Kit and their Robot.

Competitors will be allowed 3D-printed parts and parts made from sheet stock as per the requirements specified in this document.

An assembled robot must begin any test run with the following size restrictions:

The Competitor robot will be measured using a measuring device to ensure that its unique starting (pre-match set-up) configuration fits within 600 mm x 600 mm x 600 mm requirement.

Surprise Test Project Modules or Tasks – the Surprise Test Project Modules or Tasks will be released to Competitors during meetings, and no notes may be taken at this time.

3.2 Activities to be completed before the Competition

Design, Prototyping, and Testing of the Competitor Robot and manufacture of all permissible Competitor-designed components needed by the team to assemble their robot for the Test Project.

Competitors could also prepare custom wiring harnesses for the competition robot for the main Test Project.

Preparation of all items for Module B.

Note: Studica items/components are not to be modified in any way before the Competition.

4. Equipment, machinery, installations, and materials required

4.1 Equipment provided by Competition Organizer

Sample test elements,

All equipment, tools, and materials are on the Infrastructure List

- All materials for any surprise Test Project.
- Objects required for manipulation during the performance tasks and testing.
- Test court for the performance evaluation tasks.
- Test walls and objects for the navigation and localization tasks, ranging from 300 mm to 600 mm.
- Parts and tools required to complete the Prototyping module.

4.2 Equipment and material not permitted

- Any additional laptops or portable computers.
- PDA e.g. Palm, IPAQ etc.
- Memory sticks/MP3 Player/Digital Storage (except on C-2).
- Walkman radio/CD Player.
- Electronic organizer/diaries.
- Wireless communication devices including mobile phones.
- Any additional software not supplied by the Competition Organizer unless approved by Experts.
- Purchased items modified in any way before the Competition.
- Equipment that is similar or operates similarly to supplied equipment. For example – if a metal cut-off saw is provided by the Competition Organizer, then no team may provide their own metal cut-off saw.
- All raw materials, components, parts, purchased items, tools, and equipment are not permitted in the Technical Description and List of Materials Allowed and Not Allowed of the Skill.

The Skill Management Team could decide that additional items are not permitted before or during the competition, if the information is provided to Competitors/Experts.

4.3 Items to be provided by the teams

Competitors may manufacture and create custom components for their mobile robots. Some restrictions will be enforced and checked before the Competitors may use their robot on C1, independently of which module(s) those components be used being main components or spare ones.

A. All 3D printed elements must be created using ABS, PLA, Nylon, PETG, HIPS, ASA, TPU, or Carbon Filled Fiber.
B. All sheet components must be created using any polycarbonate material.
C. Competitors can bring custom cabling and electrical wires required for the robot's wiring. Custom PCBs are permitted only for custom wiring. The custom PCB may only contain headers/sockets and capacitors/resistors for any extra filtering.
D. The maximum overall weight of 3D printed elements (including both main and spare components) should be below 1.5 kg.
E. The maximum overall size of sheet components must not exceed 1000 mm x 1000 mm x 4 mm.
F. Competitors must bring three cloned copies of the Micro SD Card used on the VMX. The restriction of no Internet on-site means all packages and software must be installed beforehand.
G. Competitors must bring any custom components with them to the Competition. They will not be provided on-site.

H. On C1, Experts will inspect Competitors' custom components to ensure compliance with these restrictions. Any components outside the compliance must be modified before Competitors can use their robot.
I. Personal protective equipment.

Additionally, Competitors could bring:

- Electrical and electronic wiring harness components.
- Jigs, fixtures, formers, and clamping devices.
- Other specific manufacturing equipment or tools that are required, that is not in the Infrastructure List.

During the competition duration, no tools, equipment, stationery, components, parts, raw materials, manuals, drawings, electrical devices, or digital storage devices, unless it is approved by the SMT may be removed from or brought into the Competition venue. To enter any additional items into the competition after Familiarization Day, the approval of the SMT is necessary, and if approved, it must be checked immediately by the Experts responsible.

5. Main Test Project

The main Test Project is divided into various modules.

The Emirates Skills Autonomous Mobile Robotics Test Project requires Competitors to design, build, and operate a Mobile Robot capable of addressing ALL the performance requirements.

Competitors are expected to identify the Primary Set of Mobile Robot Performance Requirements by analyzing the information provided in this document.

5.1 Module A - Work Organization, Management and Communication

5.1.1 Completion Time

The first Module addresses sections 1 and 2 of the WSOS – “Work Organization and Management” and “Communication and Interpersonal Skills”.

Module A – Work Organization, Management, and Communication is observed throughout the competition.

5.1.2 Assessment Criteria

Work organization and management are important to the regular tasks teams are expected to perform in a cooperative environment. Moreover, there are multiple ways for a competitor to communicate and it is expected that a competitor will always communicate professionally.

The following are examples of potential criteria:

- Adherence to schedules.
- Co-operative Behavior with your Compatriot Competitor, Opponent Competitors, and the Experts.
- Workspace Organization.
- Robot Assembly Completed on Time.
- Asking before leaving the Competitor’s work area.
- Robot communication lights correctly demonstrate whether the robot is moving or not.

5.1.3 Module Task

No specific task will be done for this module, as it happens during the entire competition. During C-1, warnings could be given, but no point deduction will happen.

5.2 Module B – Design

5.2.1 Completion Time

Module B is based on an inspection of the Competitor's Assembled Robot by a panel of three Experts and the completion of some simple core tasks focusing on the following sample areas/items.

ASPECT ID	DESCRIPTION
1	Base Robot – Structural Elements
2	Wiring Installation
3	Drive System Installation
4	Object Management Design and Wiring
5	Driving Performance
6	Accuracy
7	Maintenance Test
8	Safety
9	Operation of Object Management System

5.2.2 Assessment Criteria

Module B is marked by the Experts during the competition. Information on the specific tasks and tests required will be provided to Competitors in the morning briefing.

5.2.3 Module Task

Appendix B contains the inspection and criteria on which the Experts will mark the Competitor's robot design.

5.3 Module D – Navigation and Localization

The navigation and localization module focuses on the Competitor's skill to navigate and traverse an unknown area.

5.3.1 Completion Time

Module D – Navigation and Localization Module is expected to be completed in a total 3-hour time block. After the briefing, following a random order, at least 1 hour will be given for teams to strategize about their test run, modify the code, and practice with the robot (outside the arena). Each team will have 15 minutes in the navigation arena to run the evaluation. If a team completes the task or preparation, they could use the remaining time to continue preparing for other modules, either in their workstation or the arenas.

5.3.2 Assessment Criteria

Module D is marked objectively by a panel of three Experts.

The robot can rerun to get the highest score within 600 second. The score marked will not reset to zero score when rerun.

5.3.3 Module Task

Appendix D contains an example task the Competitors can expect to see at the competition.

5.4 Module E - Vision

The vision module focuses on the Competitor's skill to create machine vision algorithms to detect and analyse different objects. The vision module is unique, as a robot is not required. However, Competitors may use their robots if they wish. An example task for the vision module is in Appendix E.

5.4.1 Evaluation Time:

Module E – Vision is expected to be completed in 3 hours.

5.4.2 Assessment Criteria:

The competitor must place the object within the designated area for 15 seconds. The result must appear before the 10th second and must remain stable for at least 5 seconds.

Module E is marked objectively by a panel of three Experts.

5.4.3 Equipment:

Competitors will be given a set of unknown objects to complete the task. From the set of objects, a random sample will be chosen and used for evaluation. The set of objects used will be the same for all competitors.

5.4.4 Module Task:

Appendix E contains an example task the Competitors can expect to see at the competition.

5.5 Module F – Robot Environment Interaction

A robot interacting with objects or a human part of its environment through physical contact must apply appropriate forces and torques. It must also adapt them to conditions that change with time safely and stably. For example, a robot breaking an egg must adjust its behavior so as not to damage the egg inside. Similarly, a robot applying force on a human arm must be able to maintain this force while accommodating any unplanned movement. This challenge is at the core of industrial applications or human-machine interaction.

5.5.1 Evaluation Time

Module F – Robot Environment Interaction is expected to be completed in a total 3-hour time block. After the briefing, at least 1 hour will be given for teams to strategize about their test run and prepare themselves. After that, following a random order, each team will have 15 minutes in the court to run the evaluation. If a team completes the task or preparation, they could use the remaining time to continue preparing for other modules, either in their workstation or the arenas.

5.5.2 Assessment Criteria

Module F is marked objectively by a panel of three Experts.

5.5.3 Module Task

Appendix E contains an example task the Competitors can expect to see at the competition.

5.5.4 Equipment

Competitors will be given an unknown object(s) that will need to be manipulated by or will interact with the robot. The basic size and weight requirements of these object(s) will be provided in advance via the official communication channel.

5.6 Module G – Performance and Commissioning

Module G is the main task of the Autonomous Mobile Robotics Skill and is based on the sample task known to all Competitors. Core performance tasks and test runs are completed in this module on a court measuring around 2m x 4m, 4m x 4m, 4m x 8m or 8m x 8m. The court layout and test objects to be used will be made available to Competitors on C1. Information on the actual requirements for the test runs will be announced during the morning briefing on C3.

5.6.1 Completion Time

Module G – Performance and Commissioning is expected to be completed at the end of C3.

5.6.2 Assessment Criteria

Performance Run 1 and 2 – Marked by Experts using a marking sheet at the end of a run.

When a robot get stuck, Competitors can reset and move their robot to a known location using tele-op(joystick) or manually and start autonomously again. However, marks will be deducted from the team's run for any adjustment. Tele-op adjustments will have smaller deductions than manually moving the robot to simulate a real event.

The robot can rerun to get the highest score within 600 second. The score marked will not reset to zero score when rerun.

5.6.3 Module Task

The task should be performed mostly autonomously, but teams are allowed to interfere with the robot (with penalties), either using teleoperation or manually (physically touching the robot). This will allow the Competitors to perform small adjustments to the robot, backing it up into a known stage. The details of this will be provided on the Test Project, but teams will be required to announce each interaction with the Experts marking that run.

5.6.4 Equipment

Module G is where the competitors will take the most advantage of the robots they designed at home. The robots will be built from the Studica Kits and the manufactured custom components.

5.7 Mock Schedule

Competitor Schedule				
Time	C-1	C1	C2	C3
Morning	Familiarization Day	Module D/E	Module F	Module G
Afternoon		Module D/E	Module G	Module G

If Competitors finish the modules early, they may use the time to practice for other modules.

Competitors can modify (build) parts of their robot during the Competition days.

5.8 Test Project Summary

Module	Days	Main Location	Equipment	Evaluation
A/B	C1 – C3	All	N/A	Judgment
D	C1	Navigation Court	Dedicated	Measurement
E	C1	Workspace	Vision Objects	Measurement
F	C2	Competition Arena	Any	Measurement
G	C2 – C3	Competition Arena	Any	Measurement

Appendix A - Work Organization, Management and Communication

No further information at this point.

Appendix B - Design

The Experts mark the design based on how well the Competitors build the robot. To help the Experts mark and understand the robot Competitors can create a 3-minute video going over the inspection list and describing what each major part of the robot does. This video should be submitted on the C1 morning period identified by their team/school code.

Appendix D - Navigation and Localization

D.1 – Example Task for Module D

Competitors must program their robot to navigate a random environment. Competitors are not allowed access to the court except for the evaluated run. The robot must be calibrated and tested in their work area or one of the main module G courts. During an evaluation run, the robot is placed in the start position and must end in the end position. No code modification is allowed once the robot enters the test area. The run is considered ended when the robot touches any part of the end position.

Appendix E – Vision

E.1 – Example set fruits for Module E

This set will not be used in Emirates Skills 2026. It should be used as sample objects for training purposes only.

Peach	Apple	Lime
		
Garlic	**Kiwi**	**Mangosteen**
		
Chili Pepper	**Banana**	**Orange**
		

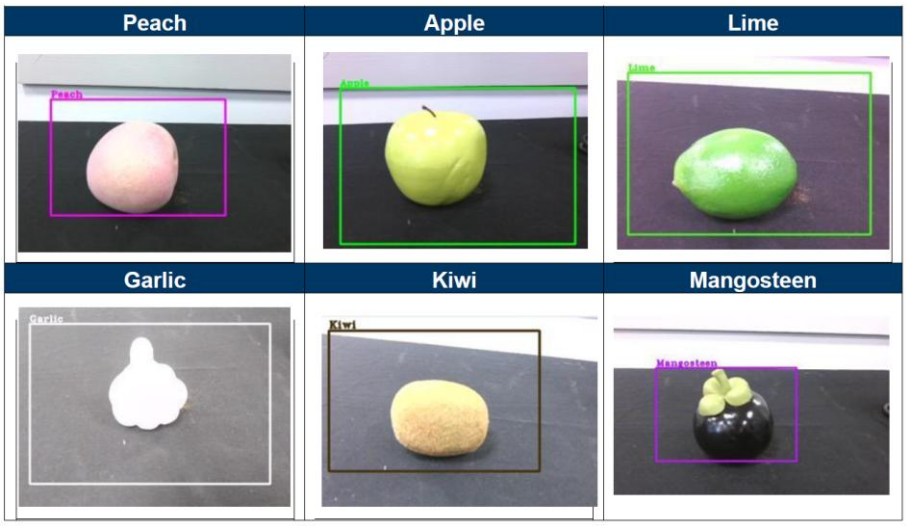
E.2 - Example Task for Module E

Five random fruits will be chosen from the fruit in E.1. The Competitors must use vision to identify the fruit.

Competitors will be marked on a shuffle board display of the objects name. The Expert will place the fruit in the camera's FOV (Field of View), and the vision programme will display the fruit's identity on the shuffle board on a dedicated laptop.

Fruit will only be placed one at a time in the camera FOV.

Sample results:



Appendix F - Robot Environment Interaction

F.1 – Example Task for Module F

Competitors will be given an unknown object for the Competitor robot to manipulate.

For the example task, Competitors will be given the Chili Pepper presented on the right. Competitor robots are required to pick up the Chili Pepper, drive 1m, turn around, drive back, and place the Chili Pepper back in the same spot it was picked up from.



In this example, the height and placement of the Chili-Pepper at the start is up to the Competitor.

Additional details will be provided during the Competition.

Appendix G - Performance Review and Commissioning

G.1 – Task Procedures

In the morning briefing, Competitors will be provided a list of possible orders to be completed. Before starting a run, a dice roll will select which order to use. The competitors will input the order in which the robot must complete their tasks directly on their computers at the beginning of the performance run.

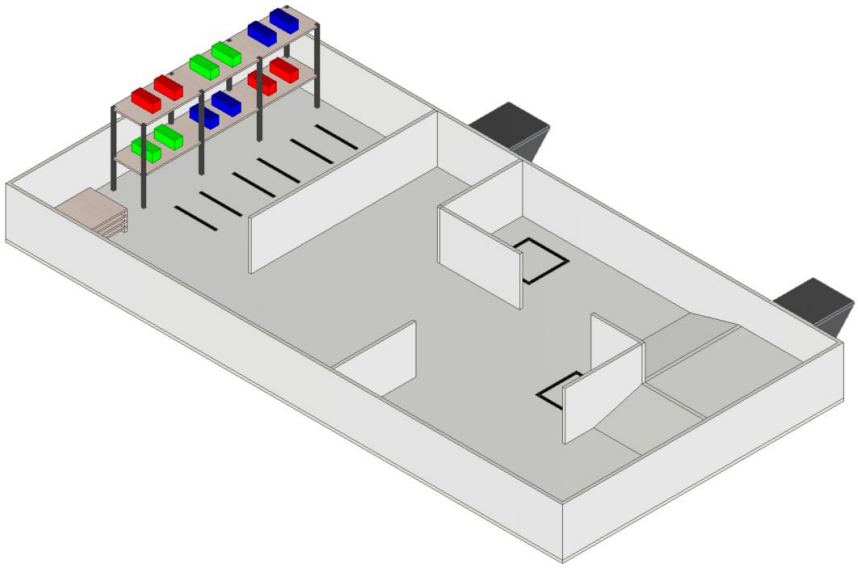
If a robot becomes stuck during a performance run, Competitors have two options to resolve the situation and resume the autonomous operation. Firstly, they may physically enter the court, relocate the robot to a predetermined position, and reactivate its autonomous mode.

Alternatively, Competitors can remotely control the robot to move it to a known location and then switch it back to autonomous mode. If remote control is employed, the run will be evaluated with a penalty, whereas if a competitor enters the court, the penalty would be higher.

Note: The Competitor's robot can complete the order using any strategy they wish. A robot can carry multiple objects at once if desired.

G.2 – Example Field for Performance Review

There are 2 runs to be completed on the example court. Runs 1 and 2 are the same, but the order and layout of the objects are different.



G.2.1 Sample Sequence for Runs 1

1. Robot is placed on the court on the Start Position.
2. Competitors place the Products on the Shelves.
3. Competitor starts the robot by hitting the start button on the robot’s control panel.
4. Robot goes out and fulfills the orders by repeatedly picking up selected Products and depositing them in the delivery places.
5. Robot goes back to the Start Position.
6. Robot flashes the red stopped LED on the control panel to indicate run complete. (Experts will not stop the time until this is displayed.)
7. Competitors return the court to its original setup.

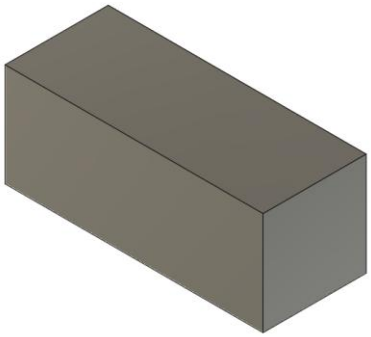
G.2.2 Sample Sequence for Runs 2

1. Robot is placed on the court on the Start Position.
2. Competitor sends the order to the robot.
3. Competitors place the Products on the Shelves.
4. Competitor starts the robot by hitting the start button on the robot's control panel.
5. Robot goes out and fulfills the orders by repeatedly picking up selected Products and depositing them in the delivery places.
6. Robot goes back to the Start Position.
7. Robot flashes the red stopped LED on the control panel to indicate run complete. (Experts will not stop the time until this is displayed.)
8. Competitors return the court to its original setup.

G.3 –Module Items

This document contains objects and elements that will be part of the final Test Project for Emirates Skills 2026 and others that are just examples.

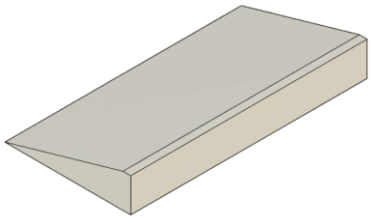
G.3.1 Competition Products

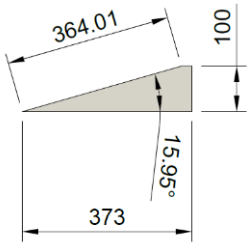


For this Emirates Skills Competition the object used will be a delivery box that has to be taken from Product Source Shelves and delivered to one of the 4 Delivery places.

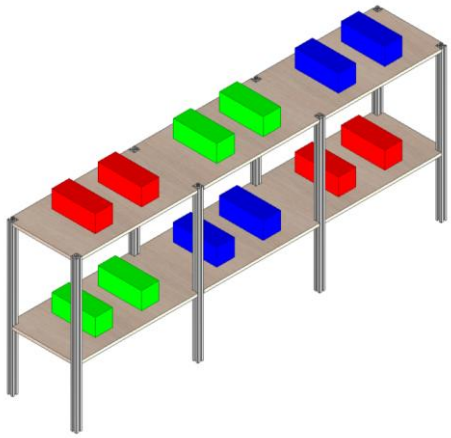
The box is 6 x 6 x 15 cm and can have 3 colors: red, green and blue.

G.3.2 Elevated Platform / Ramp





G.3.3 Product Source Shelves



The products start on the Product Source Shelves, a 2-shelf dispensary that can hold up to 6 sets of products.

Specs:

- Length: 120cm – 130cm
- First Shelf Height: 15cm – 25cm
- Second Shelf Height: 45cm – 55cm
- Depth: 20cm

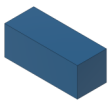
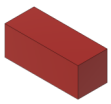
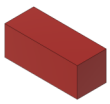
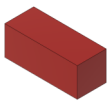
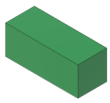
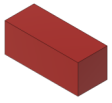
G.3.4 Pallet

A stack of 4 pallets will be provided in the court. This resource can be used to carry multiple products at the same time to the delivery position. Using a pallet is optional and the robot can carry multiple objects by itself without it.



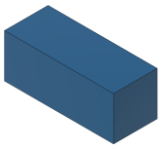
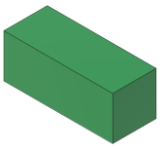
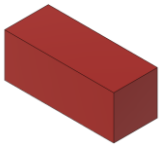
G.4 – Competition Orders

Below there is the list of possible orders that will be used in the competition.

Products / Task	1	2	3	4	5	6
Product 1						
Product 2						
Product 3						

G.5 – Product Recognition

In front of each product section there will be a tag indicating the type of product there. This tag has two different types of code to detect the product: Aruco Marker and Barcode (Code 11).

Product	ARUCO	Barcode
	 0	 023261
	 1	 023262
	 2	 023263

During the Module G Run 2 evaluation time a surprise dispensary layout will be choose and the teams have the following options to detect where each product will be:

- Read the Object color
- Read the ARUCO code in front of the shelf
- Read the Barcode in front of the shelf
- Input the layout info in the code